
REACT: Hand Pose Tracking and Regression from sEMG via User-Adaptive Conditioning

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Abstract

Surface electromyography (sEMG) enables continuous hand pose estimation on wearable devices, but models trained on multi-user corpora degrade on unseen individuals due to inter-user variability in anatomy and electrode placement. We propose **REACT**, a lightweight conditioning framework that personalizes a frozen pretrained EMG-to-pose backbone at inference time using only a handful of calibration recordings. REACT learns a compact user embedding from calibration data and applies Feature-wise Linear Modulation (FiLM) to adapt the shared encoder’s feature space, requiring no gradient updates at deployment. On the large-scale EMG2POSE benchmark, REACT improves over the state-of-the-art baseline across all three generalization splits in both regression and tracking modes, reducing angular error by up to 3.9% with minimal parameter overhead and under 45 seconds of per-user calibration.

1 Introduction

Continuous hand pose estimation from surface electromyography (sEMG) enables next-generation wearable interfaces for virtual and augmented reality, prosthetic control, and silent speech recognition [Salter et al., 2024, CTRL-labs at Reality Labs et al., 2024]. Unlike camera-based systems, sEMG-based approaches do not require line-of-sight and can operate in uncontrolled lighting, making them attractive for always-on wrist-worn devices. Recent large-scale efforts, notably EMG2POSE [Salter et al., 2024], have collected hundreds of hours of synchronized EMG–pose data across nearly 200 users and demonstrated that deep temporal models can map multichannel sEMG to continuous hand joint angles with reasonable accuracy.

However, a persistent obstacle is *inter-user variability*. Differences in muscle anatomy, subcutaneous fat, skin impedance, and electrode placement cause the same gesture to produce substantially different sEMG signatures across individuals [Atzori et al., 2016a, Phinyomark and Scheme, 2018]. Models trained on a large population degrade significantly when evaluated on held-out users or novel gesture stages [Salter et al., 2024]. Fine-tuning on per-user data is effective but expensive and impractical for consumer deployment. Meta-learning approaches such as MAML [Finn et al., 2017] address few-shot adaptation but impose high computational overhead from second-order gradients.

We propose **REACT**, a framework that bridges the cross-user generalization gap through lightweight conditional adaptation. Given k short calibration recordings from a new user (≤ 45 seconds total), REACT generates a compact user embedding that drives Feature-wise Linear Modulation (FiLM) [Perez et al., 2018] of the encoder feature space, effectively personalizing a frozen pretrained backbone without modifying its weights at inference time. Our contributions are:

1. **A modular user-adaptive architecture** that inserts between any frozen encoder–decoder pair, requiring no architectural changes to the backbone.

2. **A four-stage conditioning pathway** combining characteristic convolutions, bidirectional GRU-based temporal pooling, transformer-based set aggregation, and FiLM projection to distill variable-length, variable-count calibration recordings into user-specific feature modulation.
3. **Comprehensive evaluation** on the EMG2POSE benchmark across three orthogonal generalization axes and two prediction modes, demonstrating consistent improvements with minimal parameter overhead.

2 Related Work

EMG-based hand pose estimation. Early approaches relied on hand-crafted time- and frequency-domain features fed to classifiers [Phinyomark et al., 2012, Atzori et al., 2016a]. Deep learning methods have since dominated, with convolutional [Atzori et al., 2016b], recurrent [CTRL-labs at Reality Labs et al., 2024], and transformer-based [Rahimian et al., 2021] architectures achieving strong within-user performance. The EMG2POSE benchmark [Salter et al., 2024] introduced a large-scale dataset with three supervised baselines (VEMG2POSE, NeuroPose, and SensingDynamics) evaluated across user, stage, and user+stage generalization splits. While VEMG2POSE achieves the best baseline accuracy, all models degrade substantially on unseen users, motivating user-adaptive solutions.

User adaptation and personalization. Transfer learning and fine-tuning are widely used for EMG personalization [Côté-Allard et al., 2019, Du et al., 2017] but require gradient updates at deployment. Domain adaptation methods align source and target distributions [Ganin et al., 2016] but typically assume access to labeled target data. More recently, CPEP [Cui et al., 2025] proposed contrastive pose-EMG pre-training for zero-shot gesture classification but did not address continuous pose regression or explicit user conditioning. Our approach differs by (i) operating in the continuous pose regression setting, (ii) performing test-time adaptation without gradient updates, and (iii) using explicit user embeddings via FiLM rather than implicit alignment.

Feature-wise Linear Modulation. FiLM [Perez et al., 2018] applies learned affine transformations $\gamma \odot \mathbf{f} + \beta$ conditioned on auxiliary information, originally proposed for visual question answering. It has since been adopted for style transfer [Dumoulin et al., 2017], few-shot learning [Oreshkin et al., 2018], multi-task learning [Strezoski et al., 2019], and speaker adaptation in speech recognition [Zhao et al., 2019]. We adapt FiLM for user-specific conditioning in the biosignal domain, where it provides a natural mechanism for modulating shared representations with per-user characteristics.

Temporal pooling and set aggregation. Aggregating variable-length sequences and variable-size sets into fixed representations are fundamental challenges in our pipeline. Gated Recurrent Units (GRUs) [Cho et al., 2014] provide effective temporal summarization by maintaining a hidden state across variable-length sequences; we use a bidirectional GRU to pool per-recording features, as it captures ordering information that attention-based pooling discards. For aggregating multiple recordings, Set Transformer [Lee et al., 2019] introduced attention-based pooling with learnable query tokens; our transformer group encoder draws on this idea, using a learnable [CLS]-style query to aggregate k recording-level embeddings into a single user representation.

3 Method

3.1 Problem formulation

Let $\mathbf{x} \in \mathbb{R}^{16 \times L}$ denote a 16-channel sEMG recording of length L at 2 kHz, and $\mathbf{y} \in \mathbb{R}^{20 \times T}$ the corresponding 20-DOF hand joint angles at $T = 50$ Hz. For user u , we additionally have access to a calibration set $\mathcal{C}_u = \{\mathbf{c}_1^u, \dots, \mathbf{c}_k^u\}$ of k short recordings from other sessions of the same user. The goal is to learn a mapping $f(\mathbf{x}; \mathcal{C}_u) \rightarrow \hat{\mathbf{y}}$ that adapts to user u at inference time using only $\mathcal{C}_u$, without gradient updates.

3.2 Architecture overview

REACT inserts a conditioning pathway (~ 890 K params) between the pretrained VEMG2POSE encoder and decoder (~ 5.9 M params combined), as shown in Figure 1.

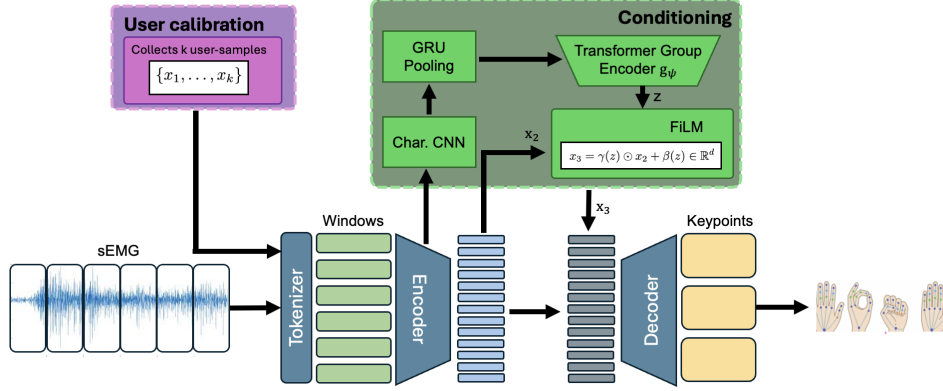


Figure 1: REACT architecture. A pretrained TDS encoder maps raw EMG to temporal features. The conditioning pathway processes k calibration recordings into a user embedding that drives a FiLM layer, which applies per-channel affine modulation (γ, β) to the encoder features before the pretrained LSTM decoder.

3.3 Pretrained backbone

We build on the VEMG2POSE model [Salter et al., 2024], a TDS convolutional encoder [Hannun et al., 2019] paired with a two-layer LSTM decoder. The encoder maps raw sEMG to temporal features; the decoder supports both regression (absolute joint angle prediction) and tracking (velocity integration from an initial pose) modes. Full architectural details are in Appendix A.

3.4 Conditioning

Each calibration recording is processed through four stages to produce user-specific modulation of the encoder features:

1. A *Characteristic CNN* that extracts user-specific features from the shared encoder output.
2. A *Bidirectional GRU* [Cho et al., 2014] that pools each recording’s temporal dimension into a fixed-size vector, capturing ordering information that attention pooling discards.
3. A *Transformer Group Encoder* [Lee et al., 2019] that aggregates the k per-recording vectors into a single user embedding $\mathbf{z}$. When $k=0$ (no calibration), a learnable fallback embedding $\mathbf{z}_0$ is used.
4. A *FiLM projection* that maps the embedding to $(\gamma, \beta) \in \mathbb{R}^{64}$, applied channel-wise as $\gamma \odot \mathbf{f} + \beta$ to the encoder output, allowing user-specific rescaling without altering the pretrained feature structure.

For stable training, γ is initialized to $\mathbf{1}$ and β to $\mathbf{0}$, ensuring the model replicates the pretrained baseline at initialization. Architectural hyperparameters are provided in Appendix A.

3.5 Training procedure

Training proceeds in two phases to balance adaptation learning and backbone preservation.

Phase 1: Adaptation module training (frozen backbone). The encoder and decoder are frozen; only the conditioning pathway is trained with AdamW [Loshchilov and Hutter, 2019] and cosine annealing for 2 epochs. At each iteration, $k \sim \text{Uniform}(0, 30)$ calibration recordings are sampled from other sessions of the same user. Calibration recordings are pre-encoded offline to avoid redundant computation.

Phase 2: Joint fine-tuning (unfrozen backbone). The encoder and decoder are unfrozen with layer-wise learning rate decay for 2 additional epochs. This prevents catastrophic forgetting of pretrained representations while allowing the backbone to co-adapt with the conditioning module.

Loss function. The combined training objective is:

$$\mathcal{L} = \underbrace{\frac{1}{N} \sum_i |\hat{y}_i - y_i|}_{\text{Angular MAE}} + \lambda \underbrace{\frac{1}{N} \sum_i \|\hat{\mathbf{p}}_i - \mathbf{p}_i\|_2}_{\text{Fingertip distance}}, \quad (1)$$

where $\hat{y}_i, y_i$ are predicted and ground-truth joint angles (in radians), $\hat{\mathbf{p}}_i, \mathbf{p}_i$ are 3-D fingertip positions derived via forward kinematics, and $\lambda = 0.01$. Invalid labels (inverse-kinematics failures) are masked during loss computation.

Data augmentation. Channel rotation augmentation simulates electrode placement variability by cyclically shifting the EMG channels [Salter et al., 2024]. Instance normalization is applied per channel per recording.

4 Experiments

4.1 Dataset

We evaluate on the EMG2POSE benchmark [Salter et al., 2024], which provides synchronized sEMG and hand pose labels. Table 1 summarizes the dataset statistics.

Table 1: EMG2POSE dataset statistics.

Split	Users	Recordings	Hours
Train	158	17,136	288.9
Validation	15	1,950	32.6
Test	178 [†]	6,167	102.7
Total	193	25,253	424.2

[†]Test includes 20 held-out users plus 158 training users evaluated on unseen gesture stages.

4.2 Evaluation protocol

Following Salter et al. [2024], we evaluate across three orthogonal generalization axes:

- **User split:** Held-out users, seen gesture stages. Tests adaptation to new users.
- **Stage split:** Seen users, held-out gesture stages. Tests generalization to novel movements.
- **User+Stage split:** Both held out. The hardest generalization setting.

We report two metrics aligned with the benchmark: **angular MAE** (degrees) and **landmark distance** (mm). Both modes of the baseline are evaluated: *regression* (absolute joint angle prediction) and *tracking* (velocity integration from an initial pose). Our baseline is the unmodified pretrained VEMG2POSE [Salter et al., 2024], the current state-of-the-art on EMG2POSE. We use the publicly released checkpoint without retraining, ensuring a fair comparison: REACT builds directly on top of this backbone, so any improvement must come from the conditioning module.

4.3 Quantitative results

Table 2 presents results at calibration budget $k=15$ across all three generalization splits and both prediction modes. REACT outperforms VEMG2POSE on **all metrics in both regression and tracking**. The largest gains appear on the Stage split for regression (MAE: $15.2 \rightarrow 14.6^\circ$, 3.9% reduction) and the User split for tracking (MAE: $7.7 \rightarrow 7.4^\circ$, 3.9% reduction), where user-specific conditioning most directly addresses the domain gap.

Table 2: **Results at $k=15$ calibration recordings.** Angular error (degrees, $\downarrow$) and landmark distance (mm, $\downarrow$). Bold indicates best. REACT improves on all splits and modes.

Split	Model	Regression		Tracking	
		Err ($^\circ$)	Dist (mm)	Err ($^\circ$)	Dist (mm)
User	VEMG2POSE	12.2	15.8	7.7	10.3
	REACT (ours)	12.0	15.6	7.4	10.0
Stage	VEMG2POSE	15.2	20.4	11.2	15.2
	REACT (ours)	14.6	19.9	10.9	14.9
User+Stage	VEMG2POSE	15.8	21.6	11.0	15.4
	REACT (ours)	15.3	21.2	10.8	15.3

Tracking mode consistently outperforms regression across all models (7.4° vs. 12.0° for REACT on the User split), since velocity integration with a known initial pose provides a strong prior that constrains the solution space. This advantage comes at the cost of requiring the initial pose, which may not always be available in deployment.

4.4 Qualitative results

REACT’s pose reconstructions (Figure 2) capture fine finger articulations (e.g., the opening motion from $t=0.0$ s to $t=2.0$ s) and produce anatomically plausible meshes throughout. Per-finger analysis reveals that REACT’s improvement is most pronounced for the pinky ($11.4^\circ \rightarrow 11.1^\circ$ in tracking) and middle fingers, which exhibit the highest inter-user variability. The thumb, which has the most constrained range of motion, shows the smallest baseline error and thus the smallest absolute improvement.

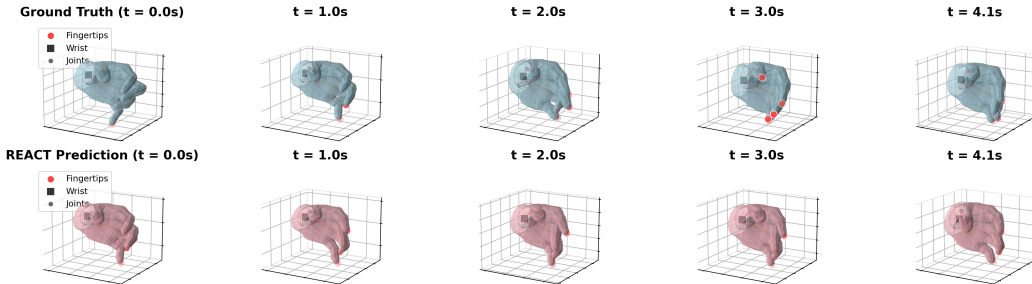


Figure 2: **Top:** ground-truth mesh and keypoints from a camera-based tracker. **Bottom:** REACT’s predicted mesh and keypoints at the same timesteps, generated from wristband EMG of a held-out user.

5 Discussion

Why does FiLM conditioning work? We hypothesize that inter-user variability manifests primarily as channel-wise scale and offset differences in the encoder feature space, arising from differences in muscle activation amplitudes and electrode impedance. FiLM’s affine transformation is well-suited to correct these per-channel distributional shifts. The identity initialization ($\gamma=1, \beta=0$) ensures that REACT replicates the pretrained baseline before learning, providing a conservative adaptation that cannot harm pre-existing performance at initialization. Despite its negligible parameter cost, the FiLM layer produces consistent improvements, suggesting that the pretrained encoder features are already expressive; they merely need user-specific rescaling to account for individual signal characteristics.

GRU pooling outperforms attention pooling. Early experiments used attention-based temporal pooling. Switching to bidirectional GRU pooling improved results, likely because GRUs capture the temporal structure within calibration recordings, specifically the ordering and dynamics of muscle

activations, rather than treating frames as an unordered set. This finding aligns with the sequential nature of sEMG signals, where temporal context is informative of user-specific motor patterns.

Regression vs. tracking. REACT’s improvements are larger in tracking mode (up to 3.9% angular MAE reduction) than regression mode (up to 3.2%). We attribute this to tracking’s recurrent velocity integration, where small per-timestep improvements compound over the prediction horizon. Regression predicts each frame more independently, limiting the compounding benefit of conditioning.

Limitations. Several limitations warrant discussion: (i) Our improvements, while consistent, are moderate (2–4%); more aggressive adaptation (e.g., adding FiLM to multiple encoder layers) may yield larger gains at the cost of increased complexity. (ii) We evaluate only on the EMG2POSE dataset; generalization to other EMG devices, electrode configurations, or clinical populations remains untested. (iii) REACT relies on labeled calibration recordings with ground-truth poses, which requires a camera-based tracking session, a practical constraint for deployment. Future work could explore self-supervised calibration using unlabeled EMG data. (iv) The current design applies FiLM at a single point in the architecture; multi-scale conditioning may capture richer user characteristics.

6 Conclusion

We presented REACT, a lightweight conditioning framework for EMG-to-pose estimation that bridges the cross-user generalization gap through Feature-wise Linear Modulation. By inserting a modular conditioning pathway between a frozen pretrained encoder and decoder, REACT personalizes predictions using only a handful of calibration recordings with minimal parameter overhead. On the EMG2POSE benchmark, REACT consistently improves angular error and landmark distance across all generalization splits in both regression and tracking modes. Our results demonstrate that conditional adaptation is a practical strategy for deploying EMG-based hand pose systems across diverse user populations, and we believe this modular approach generalizes to other biosignal domains where subject variability is a bottleneck.

Future work. Promising directions include: (i) multi-scale FiLM conditioning at multiple encoder layers, (ii) contrastive pre-training of the user encoder for richer embeddings, (iii) online adaptation during continuous use, and (iv) evaluation on clinical populations with neuromotor conditions.

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A Implementation Details

Encoder architecture. The TDS encoder consists of two strided 1-D convolution blocks followed by two TDS stages:

- Conv1dBlock: $16 \rightarrow 256$ channels, kernel size 11, stride 5
- Conv1dBlock: $256 \rightarrow 256$ channels, kernel size 5, stride 2
- TDS Stage 1: 2 blocks, in_channels=256, channels=16, feature_width=16, kernel=9
- TDS Stage 2: 2 blocks, same config with kernel=5, out_channels=64

Left context: 1,790 samples at 2 kHz (≈ 0.9 s). Input windows: 11,790 samples (10,000 content + 1,790 context), strided by 2,000 samples (1 s).

Decoder architecture. The LSTM decoder is a 2-layer LSTM with hidden size 512:

- Regression mode: in_channels=84 (64 features + 20 state), out_channels=40 (position + velocity)
- Tracking mode: in_channels=84, out_channels=20 (velocity only)
- Output scale: 0.01; operating frequency: 50 Hz

In regression mode, the first 500 samples at 2 kHz ($\equiv 12.5$ steps at 50 Hz) use direct position output, then switch to velocity integration.

User encoder details. The characteristic CNN uses kernel size 3, layer normalization, GELU activations, and dropout. The bidirectional GRU has 3 layers, hidden size 64, and dropout 0.1. The transformer group encoder uses 3 layers, 4 attention heads, and hidden dimension 128, with sinusoidal positional encodings supporting up to 32 calibration recordings. The FiLM projection MLP uses LayerNorm and GELU activations, with γ initialized to 1 and β to 0.

Training hyperparameters. Phase 1: AdamW with $\text{lr} = 10^{-3}$, weight decay = 10^{-4} , cosine annealing, and 500-step linear warmup. Phase 2: base $\text{lr} = 10^{-4}$ with decay factor 0.5 per layer depth.

Training infrastructure. All experiments were conducted on a single NVIDIA H100 GPU via Modal cloud computing. Batch size: 128. Mixed-precision training (AMP) was enabled. Total training time: ~ 4 hours for the full two-phase procedure.

B Decoder Operating Modes

The VEMG2POSE decoder supports two complementary modes:

Regression mode predicts absolute joint angles from each EMG window independently. The decoder outputs a 40-dimensional vector (20 positions + 20 velocities) at each 50 Hz timestep. For the first 12.5 steps ($= 500$ samples at 2 kHz), the position output is used directly; thereafter, velocity is integrated: $\hat{\mathbf{y}}_t = \hat{\mathbf{y}}_{t-1} + \hat{\mathbf{v}}_t$.

Tracking mode requires an initial pose (from ground truth) and predicts velocities only (20-dimensional output). Pose is recovered via pure integration: $\hat{\mathbf{y}}_t = \hat{\mathbf{y}}_{t-1} + \hat{\mathbf{v}}_t$ for all timesteps. This mode typically achieves lower absolute error since it benefits from the ground-truth anchor, but requires the initial pose to be known.